

Quantitative Researcher Superday Presentation

As part of the interview process for the Quantitative Research role at Trexquant, we are excited to invite you to deliver a short presentation to our research team. This is an opportunity to showcase how you approach and reason through an open-ended problem. **Please select one of the three topics below.**

Duration: 20-minute presentation, followed by a 10-minute Q&A with the interview panel.

Topic A: Predicting Daily Stock Returns

The Question

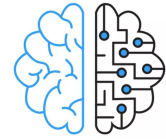
Pick **one** publicly available data source and explain how you would use it to predict **next-day stock returns** across a cross-section of liquid equities.

The target is the daily return of each stock relative to the others in your universe — i.e. which stocks will outperform tomorrow, not whether the market will go up.

Depth over breadth. We would much rather see one data source examined thoroughly than five sketched briefly. A presentation covering a single source, its quirks, and its failure modes in real depth will score higher than a survey.

What We Are Looking For

- **Economic thesis.** Why do you think this data will work? What economic or behavioral mechanism would cause this data to carry information about future returns? Who is on the other side of the trade?
- **Raw data.** Demonstrate that you have deeply understood the data itself. What does one observation look like? What is the coverage, history length, and number of stocks? When is data available relative to when events occur? What quality issues exist, and for which stocks is coverage weakest?
- **Feature engineering.** How do you turn the raw fields into a single number per stock per day? Address scaling across stocks of different size or liquidity, outliers, and missing observations.



- **Evaluation criteria.** Define your success metrics before looking at results. What would you measure, over what period, and what result would convince you the signal works? If your empirical results contradict your economic thesis, what are your next steps?
- **Limitations and next steps.** What did you not resolve? What would you do with another month?

On Empirical Results

You may spend several hours on this, but **we do not expect extensive empirical work**. We are far more interested in how you think and reason through the problem than in good backtest results.

If you do build something, be prepared to explain exactly what you did and which tools you used — we encourage you to use any resources available, including AI tools.

Some data sources candidates have explored: daily price and volume, SEC EDGAR filings, earnings announcement dates, FINRA short interest, Google Trends. *This list is illustrative — an unusual choice, well justified, is a plus.*

Format & Logistics

- You may use Google Slides or PDF. Please ensure your materials are clear, concise, and professional.
- Please send your slides at least 24 hours in advance to your recruiter.
- The presentation will be conducted in person with a few Experienced Researchers at the firm.
- As this is a superday, expect to present alongside 1–3 other candidates.
- **Expected effort: a few focused hours.**
- We look forward to your presentation and the opportunity to learn more about how you think.



Topic B: Designing an Agentic AI Alpha Researcher

The Question

Design an AI agentic loop to produce robust alpha signals — and explain how you would evaluate that system and improve it over successive iterations.

An alpha signal is a daily cross-sectional score: one number per stock per day, where the ranking across stocks predicts their relative forward returns.

What you can assume you have access to:

1. Production data features
2. Backtesting environment that allows you to test how an alpha signal performed in the past
3. Frontier LLM API access, with a finite token budget
4. A research literature corpus, if your design needs one

Additional assumptions are fine, but state them in your presentation.

What We Are Looking For

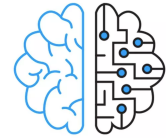
System Design

- **Task specification.** What exactly goes in and what exactly comes out? What does one finished, correct output look like? At minimum, we would like the output to have an economic thesis and the alpha signal implementation.
- **Agent Roles and Definitions.** What distinct agents would you have and their roles be? Walk through each agent and how they work together.

On Data and Effort

Do not purchase data or subscriptions for this exercise.

Prototyping is highly recommended but **not required**. A clear design with a rigorous evaluation and improvement plan scores as highly as a demo. If you do build something, show us a real example of it producing a bad signal or a bad thesis, and explain what you would change in response. We encourage you to use any resources available, including AI tools.



Expected effort: a few focused hours. Additional time will not improve your score. One narrow claim examined carefully beats a full reproduction attempted in a hurry, and a partial result you understand beats a complete one you cannot defend.

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Topic C — Reproduce, Refute, or Extend a Recent Paper

The Task

Choose one published academic paper that relates to equities, options or futures. Then structure your presentation in three parts.

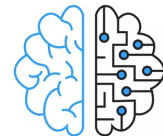
Part 1 — Overview of Paper

Explain the paper to a colleague who has not read it. What question is it asking, what is the central mechanism, what did the authors actually do, and what is the headline result? Be precise about the construction. If there is a signal, we should understand exactly how it is computed and what it is sorted against.

Part 2 — Why you chose it

What drew you to this paper? A result you found surprising, a mechanism you found convincing, a step you found suspicious, an application the authors did not consider. Tell us what you thought when you read it.

Part 3 — Your contribution (*the bulk of the presentation*)



Pick one of the following three tracks and go deep. Do not attempt more than one.

- **Refute.** Make the strongest case you can that the result will not survive. This must be a specific, testable objection, not general skepticism.
- **Reproduce.** Rebuild the central result, or a scaled-down version of it, and report what you found. A partial reproduction on a shorter sample or a smaller universe is completely acceptable — state clearly what you changed and why. If your numbers differ from the paper's, what explains the gap?
- **Extend.** Take the idea somewhere new. A different asset class, a different horizon, a different data source, a combination with something else, a mechanism test the authors did not run. Be clear about what is yours and what is theirs, and about why the extension is interesting rather than merely different.

On Data and Effort

Do not purchase data or subscriptions for this exercise.

Empirical results are welcome but **not required**. We are more interested in whether you know what to test than in whether you had time to run it.

Expected effort: a few focused hours. Additional time will not improve your score. One narrow claim examined carefully beats a full reproduction attempted in a hurry, and a partial result you understand beats a complete one you cannot defend.

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